COMP 3975
Server-Side Web Scripting

School	School of Computing and Academic Studies
Program	Computer Systems Technology (CST) Diploma - Combined Web and Mobile Option
Course Credits	5
Minimum Passing Grade	50%
Start Date	January 05, 2026
End Date	April 24, 2026
Total Hours*	75
Total Weeks	15
Hours/Weeks	5
Delivery Type	Lecture/Lab
Zero Textbook Cost	Yes
Prerequisite(s)	• Completion of first year CST and Admission into the Combined (Web/Mobile) Option
CRN	86083

Acknowledgement of Territories

The British Columbia Institute of Technology acknowledges that our campuses are located on the unceded traditional territories of the Coast Salish Nations of Sḳwx̱wú7mesh [1] (Squamish), səl̓ilwətaʔɬ [2] (Tsleil-Waututh), and xʷməθkʷəy̓əm [3] (Musqueam).

Instructor Details

Name	Medhat Elmasry
E-mail	Medhat_Elmasry@bcit.ca
Location	Online
Office Hours	Office Hours will be posted on the Learning Hub. Office hours will be via video conferencing.

Course Description

This hands-on course introduces server-side scripting using PHP (PHP: Hypertext Preprocessor) an open-source, platform-independent, server-side web scripting language. Topics include: PHP theory and syntax, textfile manipulation, cookies, regular expressions, sessions, graphics, and database interactions. By the end of this course, students have learned the major functions of PHP server-side scripting and how to freely obtain a wide variety of open-source software tools and code libraries. Students also learn how to write reusable classes and object-oriented scripts. Students will use popular PHP frameworks such as Laravel and/or Yii in addition to unit-testing frameworks such as SimpleTest or PHPUnit. Students also learn how to create secure sessions, apply hashing and encryption techniques, and use the PHP command line interface.

Course Learning Outcomes/Competencies

Upon successful completion, the student will be able to:

- Explain the theory behind server-side Web scripting.
- Write PHP Web scripts with and without HTML content.
- Identify, locate, download, and install free server-side Web scripting tools.
- Identify, locate, and download sample PHP scripts and resources relevant to PHP scripting.
- Install libraries that enable the use of resources such as compression, encryption, password security.
- Obtain, process, and store user input from Web pages.
- Develop PHP scripts that use cookies to remember users and user data.
- Develop PHP scripts that use regular expressions to find, process, and validate data.
- Develop PHP scripts that use server-side resources with their Web pages including files, databases, and images that are located on the server.
- Develop PHP scripts that use sessions to authorize users and keep session data persistent across multiple Web pages.
- Implement custom classes that generate PHP objects.
- Use PEAR or other open-source classes to generate ready-made objects.
- Discuss advanced database issues that are relevant to utilizing non-mySQL databases with PHP.
- Discuss security issues that are relevant to PHP website construction.
- Write secure and robust PHP code that is resistant to errors and attacks.
- Use PHP to generate pdf documents dynamically.
- Use Object Oriented Programming techniques in authoring PHP scripts.
- Use PHP frameworks and templating engines such as Yii, Laravel, and Smarty.
- Use unit-testing frameworks such as SimpleTest or PHPUnit.

Learning Resources

No textbook required

Evaluation Criteria

Criteria	%	Comments
Assignments	20	2 assignmenets
Midterm Exam	20	
Final Exam	20	To pass the course the average of mid-term & final exam must be at least 50%.
Project	20	
Labs	20	

Attendance Requirements

Regular attendance in lectures, seminars, and labs is seen as critical to student success, and may be monitored by faculty.

Unapproved absence of 2 or more classes may result in failure or forced withdrawal from the course or program.

Please see Policy 5101 - Student Regulations: https://www.bcit.ca/files/pdf/policies/5101.pdf [4]

Other information

Student Surveys (Instructional Assessments Online) - You'll complete an online survey near the end of this course to share feedback with your instructor. Surveys are anonymous, and results are released only after all final grades are submitted.

How to Access - Look for an email around Week 10 sent to your myBCIT email with a link to your survey. Click the link and choose the correct course from the "Active Tasks" list.

In-class time - Your instructor may provide 15 minutes of class time for completing the survey (individually).

Troubleshooting - If you don't receive the link or have issues, email bcit_feedback@bcit.ca before Week 11.

Tip - Make sure the survey you complete matches both your course CRN and your instructor.

Professionalism - Please provide honest, specific, and constructive feedback in your comments—your voice makes a difference.

Questions? Contact PDS Coordinator: Kirsty Barclay-Estrup**.**

Course Schedule and Assignments

Week	Starting	Essential Skill	Lab Exercise	Assignments
1	Jan 5	Getting Started with PHP	zodiac	
2	Jan 12	Docker & MySQL & PHP	Northwind DB	
3	Jan 19	DockerHub & Azure & Ollama & MySQL CRUD	PHP AI	Assignment #1
4	Jan 26	SQLite & State	Students CRUD SQLite	

5	Feb 2	OO & REST API	API & AI	
6	Feb 9	CSV & Chart JS & cURL	OO Regex	Assignment 1 due
7	Feb 16 (Mon Fam Day)	React.js	React	Assignment 2
8	Feb 23	Midterm Exam Week		
9	Mar 2	Laravel with MySQL	TBA	
10	Mar 9	Laravel with SQLite & API & JWT	TBA	Instructional Assessments Online
	Mar 16	Computing Reading Week		
11	Mar 23	Session Auth & Tidbits	TBA	Assignment 2 due
12	Mar 30 (Good Fri)	Azure Deploy & Swagger	TBA	
13	Apr 6 (Easter Mon)	Static Web Apps	TBA	
14	Apr 13	Study/Makeup/Review/Presentation Week	Project due + presentation	
15	Apr 20	Final Exam Week		